

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	4 MARKS

2.3 – Brainstorming

Electric vehicles are getting really popular because they are good for the environment and cost less to run. People who use electric vehicles often worry about understanding how much charge is left in the battery how far they can drive and how well the vehicle is charging. The problem is that electric vehicles produce a lot of data that's hard to understand without the right tools to visualize it. This project is about creating a way to visualize the data using Tableau so people can see how their electric vehicle is charging and how far it can go.

The system helps people see how the battery charge level affects how far they can drive, which makes it easier for them to plan their trips and not worry much about running out of charge. With the dashboards people can track how often they charge their vehicle how long it takes to charge how healthy the battery is and how much energy they use over time. The analysis can also show how things like driving style, weather and the type of vehicle affect the battery and how far it can go. People can use the charts and graphs to look at the data in different ways and get a better understanding of how their electric vehicle works.

The visualization tool helps people make decisions about their vehicle by giving them clear and easy to understand information about how it is performing. It can help people who own vehicles charge their cars at the right time use less energy and make their batteries last longer. The people who make vehicles and the researchers can use the information to see how the vehicles are doing and find ways to make them better. The people who make policies and plan the infrastructure can also use the analysis to understand how much charging is needed and build charging stations that work well. By using Tableau to visualize the data the project makes it easy for people to understand the information, about their vehicle and helps make electric vehicles more popular and transportation more sustainable.